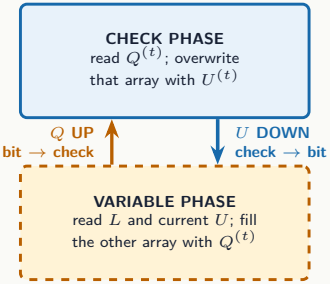
BP in one loop: Q goes up and U comes down

STATE NOW
BETWEEN ITERATIONS: the three channel LLRs stay fixed; four current edge-addressed U values are carried into the next message-passing iteration.

MEMORY NOW
fixed L: 3 current U: 4
other array[4]: available
temporary variable: none
values in use: 7 arrays: 11 positions

VARIABLE PHASE FORMS $Q^{(t)}$ ⟶ **CHECK PHASE FORMS $U^{(t)}$** ⟶ **REPEAT**



Variable-to-check rule

$$Q_{i\to c}^{(t)} = L_i + \sum_{d\in N(i)\setminus\{c\}} U_{d\to i}^{(t-1)}$$

Fixed L_i plus information from checks other than the recipient.

The two checks in this example have degree two
Each outgoing U equals the other incoming Q.

Two four-position arrays exchange roles. One holds current U; the other holds Q and is then overwritten by new U. A temporary variable performs each degree-two pair swap.

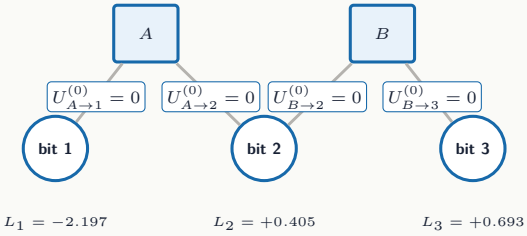
Memory: $L[3]$ + two arrays$[4+4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded

Initialize once: channel LLRs and four edge addresses

STATE NOW
INITIAL VALUES: fixed $\mathbf{L} = (-2.197, +0.405, +0.693)$ and current $\mathbf{U}^{(0)} = (0, 0, 0, 0)$. No Q value has been formed yet.

MEMORY NOW
fixed L: 3 current U: 4
other array[4]: available
temporary variable: none
values in use: 7 arrays: 11 positions

PLACE THREE FIXED CHANNEL LLRS + INITIALIZE FOUR CURRENT U VALUES



$$L_i = \log \frac{p(y_i \mid b_i = 1)}{p(y_i \mid b_i = 0)}$$

Positive L_i favours bit 1 under this deck's sign convention.

ELEVEN ARRAY POSITIONS; SEVEN VALUES IN USE NOW

$L[3]$	fixed throughout the decode
current[4] $= U^{(0)}$	carried forward
other[4]	available; next $Q^{(1)}$, then $U^{(1)}$
temporary variable	used only while converting Q to U

array positions: 11 **check-phase peak: 12**

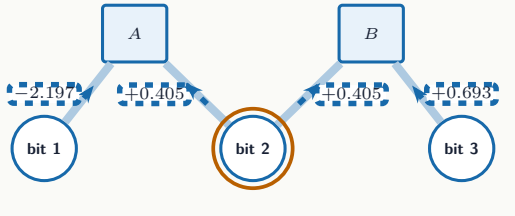
Memory: $L[3]$ + two arrays$[4 + 4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded

Variable phase: all four $Q^{(1)}$ messages rise together

STATE NOW
FROZEN READ: all four $Q^{(1)}$ values use the same current $\mathbf{U}^{(0)}$ and fill the other array. Checks A and B are shown together; neither consumes the other's new result.

MEMORY NOW
fixed L: 3 current U: 4
other array$[4]$: $Q^{(1)}[4]$
temporary variable: none
values in use: 11 arrays: 11 positions

READ FIXED L AND CURRENT $U^{(0)}$ $\longrightarrow$ FILL THE OTHER ARRAY$[4]$ WITH $Q^{(1)}$



$$Q^{(1)}_{i\to c} = L_i + \sum_{d\in N(i)\setminus\{c\}} U^{(0)}_{d\to i}$$

$$Q^{(1)}_{1\to A} = -2.197, \quad Q^{(1)}_{2\to A} = +0.405,$$
$$Q^{(1)}_{2\to B} = +0.405, \quad Q^{(1)}_{3\to B} = +0.693.$$

Degree-one bits have no other check:
$Q^{(t)}_{1\to A} = L_1$ **and** $Q^{(t)}_{3\to B} = L_3$ **for every** t.

Every outgoing Q begins with the fixed L_i and adds only current U values from checks other than its recipient.

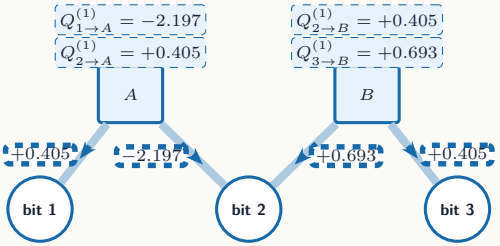
Memory: $L[3]$ + two arrays$[4+4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded

Check phase: both checks form all four $U^{(1)}$ messages

STATE NOW
ONE CHECK-PHASE SNAPSHOT: the other array changes from $Q^{(1)}[4]$ to $U^{(1)}[4]$. Each degree-two pair is read before those same two positions are overwritten.

MEMORY NOW
fixed L: 3 current U: 4
other array[4]: $Q^{(1)} \to U^{(1)}$
temporary variable: 1 during conversion
values in use: 12 peak arrays: 11 positions

READ ONE Q PAIR ⟶ **USE ONE TEMPORARY VARIABLE** ⟶ **OVERWRITE THE SAME TWO POSITIONS WITH U**



DEGREE TWO: EACH OUTGOING U IS THE OTHER INCOMING Q

check A

$$U^{(1)}_{A\to 1} = +0.405,$$
$$U^{(1)}_{A\to 2} = -2.197$$

check B

$$U^{(1)}_{B\to 2} = +0.693,$$
$$U^{(1)}_{B\to 3} = +0.405$$

Same four positions; new quantities

$[-2.197, +0.405 \mid +0.405, +0.693]_Q$
⇓ swap each degree-two pair using one temporary variable
$[+0.405, -2.197 \mid +0.693, +0.405]_U$
Q and U are different messages; only the positions are reused.
After both checks, exchange the two array roles.

Memory: $L[3]$ + two arrays$[4+4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded

Λ includes every neighboring check; outgoing Q excludes its recipient

STATE NOW
AFTER EXCHANGING ARRAY ROLES: $\mathbf{U^{(1)}}$ is current. The other array temporarily holds the three $\Lambda^{(1)}$ values; one position remains available. The L_i values stay fixed.

MEMORY NOW
fixed L: 3 current U: 4
other array[4]: $\Lambda^{(1)}[3] + 1$ free
temporary variable: none
values in use: 10 arrays: 11 positions

DECISION VALUE: INCLUDE EVERY NEIGHBOR $\longrightarrow$ **OUTGOING Q: EXCLUDE THE RECIPIENT CHECK**

DECISION VALUE: ALL NEIGHBORS

$$\Lambda_i^{(1)} = L_i + \sum_{c \in N(i)} U_{c \to i}^{(1)}$$

Includes every check joined to bit i. It is used for the current decision; the fixed L_i remains unchanged.

OUTGOING TO c: OTHER CHECKS ONLY

$$Q_{i \to c}^{(2)} = L_i + \sum_{d \in N(i) \setminus \{c\}} U_{d \to i}^{(1)}$$

$$Q_{i \to c}^{(2)} = \Lambda_i^{(1)} - U_{c \to i}^{(1)}$$

NO OTHER CHECK REMAINS — bit 1 sends to A
$N(1) = \{A\}$, so $N(1) \setminus \{A\} = \varnothing$.

$$Q_{1 \to A}^{(2)} = L_1 + 0 = L_1 = -2.197$$

CHECK B REMAINS — bit 2 sends to A
$N(2) = \{A, B\}$, so $N(2) \setminus \{A\} = \{B\}$.

$$Q_{2 \to A}^{(2)} = L_2 + U_{B \to 2}^{(1)} = +1.099$$

If no other check remains, the index set is empty and the summation contributes zero. If another check remains, its U value is included.

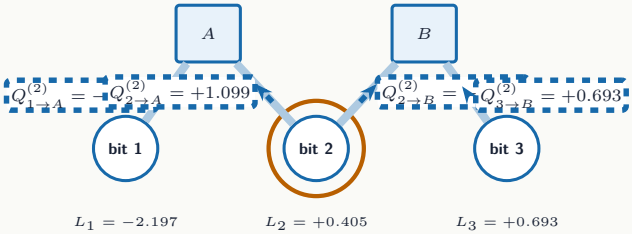
Memory: $L[3]$ + two arrays$[4 + 4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded

Second variable phase: one bit sends two recipient-specific values

STATE NOW
FROZEN READ: fixed $\mathbf{L}$ and current $\mathbf{U}^{(1)}$ fill the other array with all four $Q^{(2)}$ values.

MEMORY NOW
fixed L: 3 current U: 4
other array[4]: $Q^{(2)}[4]$
temporary variable: none
values in use: 11 arrays: 11 positions

SAME BIT AND SAME L_2 + DIFFERENT RECIPIENT EXCLUSION = DIFFERENT Q



to check A: remove A's preceding contribution

$$Q^{(2)}_{2\to A} = \Lambda^{(1)}_2 - U^{(1)}_{A\to 2} = +1.099$$

to check B: remove B's preceding contribution

$$Q^{(2)}_{2\to B} = \Lambda^{(1)}_2 - U^{(1)}_{B\to 2} = -1.792$$

$Q^{(2)}_{2\to A} \neq Q^{(2)}_{2\to B}$ **even though both originate at bit 2.**

The edge address determines which preceding check-to-variable value is excluded.

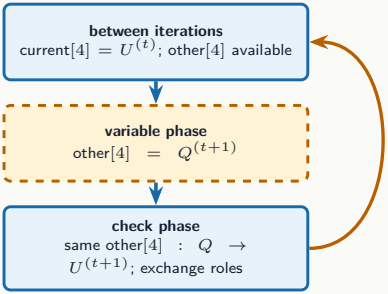
Memory: $L[3]$ + two arrays$[4+4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded

What remains between iterations, and how the two arrays exchange roles

STATE NOW
BETWEEN ITERATIONS: the three L values remain fixed and four current $U^{(t)}$ values are carried forward. Q and Λ do not need to remain.

MEMORY NOW
fixed L: 3 current U: 4
other array$[4]$: available
temporary variable: none
values in use: 7 arrays: 11 positions

FILL THE OTHER ARRAY WITH Q ⟶ OVERWRITE IT WITH NEW U ⟶ EXCHANGE ARRAY ROLES



Between phases, the other array may hold the three Λ values used for the current decision; it is then reused for the next $Q[4]$.

ARRAY POSITIONS KEPT THROUGHOUT THE DECODE

$L[3]$	3 fixed number positions
array A$[4]$	4 positions
array B$[4]$	4 positions
A and B exchange current/other roles	
total array positions	**11**

VALUES IN USE AT EACH STAGE
between iterations: $3L + 4U = 7$
while the other array holds Q: $3 + 4 + 4 = 11$
one temporary variable during a degree-two pair conversion
maximum in use: $3 + 4 + 4 + 1 = 12$
optional hard-decision estimate: 3 bits, counted separately
this degree-two reuse schedule is not a universal BP memory bound

Memory: $L[3]$ + two arrays$[4 + 4] = 11$ positions; slide 4 briefly uses one temporary variable (peak 12); graph/address data and hard decisions excluded